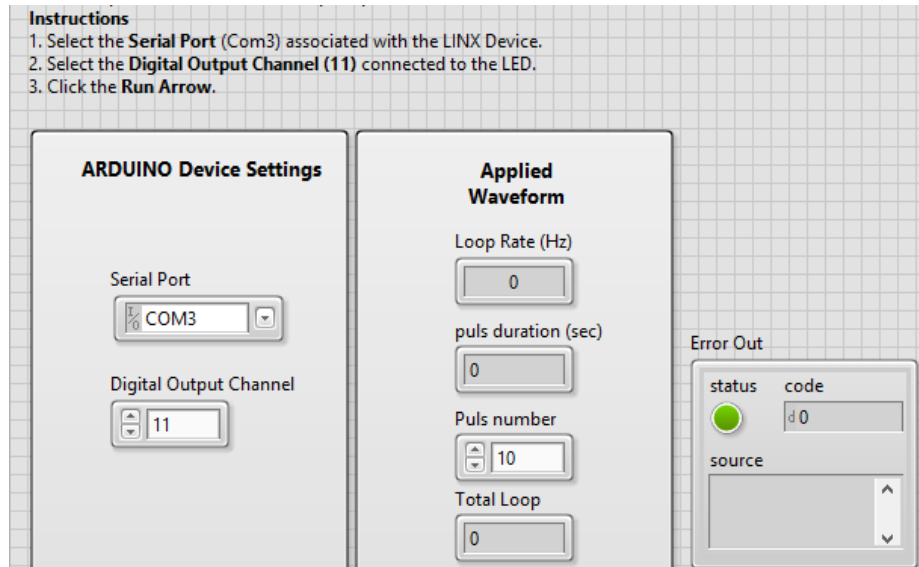


In-house micro-electroporation system and Protocol

Part 1: Electroporation system: Software installation and hardware setup

1. NI instruments are used in this experiment. The following are the required softwares and instruments:
 - a. LabView (Any version after 2013)
 - b. Arduino UNO an electronic chip for square wave generation (Simple arduino system)
 - c. Electronic chopper circuit (Transistor circuit)
 - d. Power source (5V-20V)
 - e. Platinum electrodes
2. To install LabView, an account must be created and the license must be purchased. LabView can be purchased from the following NI website and automatically updates the versions for valid licenses. <https://www.ni.com/en-tr/shop/labview.html>
3. Also ensure that the Arduino software is also installed on the device
4. To connect the arduino to LabView, an arduino software must be installed. The following is a sample video showing an example for Arduino pairing with LabView. (<https://www.youtube.com/watch?v=RGRhIQneO6w>)
5. LabView requires an extension to enable the programming of the Arduino. The extension can be obtained from the software installed from the following website: <https://www.vipm.io/desktop/>
6. The software is called VI Package manager and it requires an account to allow free download of the software.
7. On VI Package manager, search for Arduino and install the latest version for the LabView interface for Arduino by double clicking it and selecting install.
8. After installation, a folder will appear in the installed LabView files called LINX. Open this file
9. Open the LabVIEWInterface.h folder and define the Arduino Baud Rate = 9600
10. Go to the base file, select the serial port and the arduino board that you are using. Run this base file and upload it on the board.
11. Once this step is done, the Arduino is ready to take signals and code from the LabView software. The Arduino program can be closed.
12. Open the LabView software and open the provided LabView file in the software. (labview plasmid injection-final-remove for loop.vi)
13. The LabView interface looks as seen in the image below:

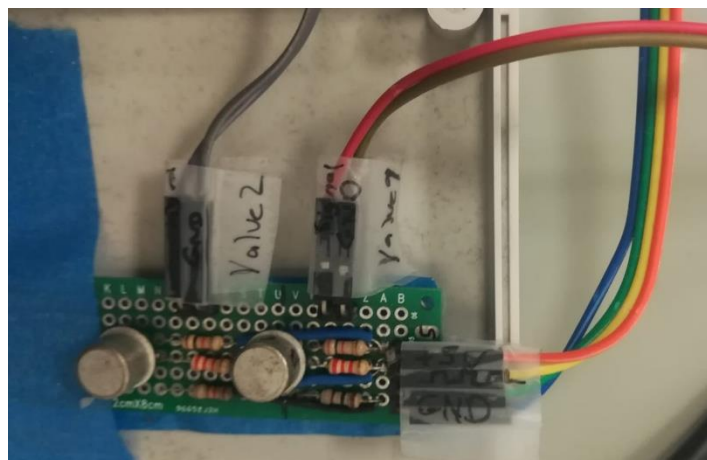


14. The software setup is complete.

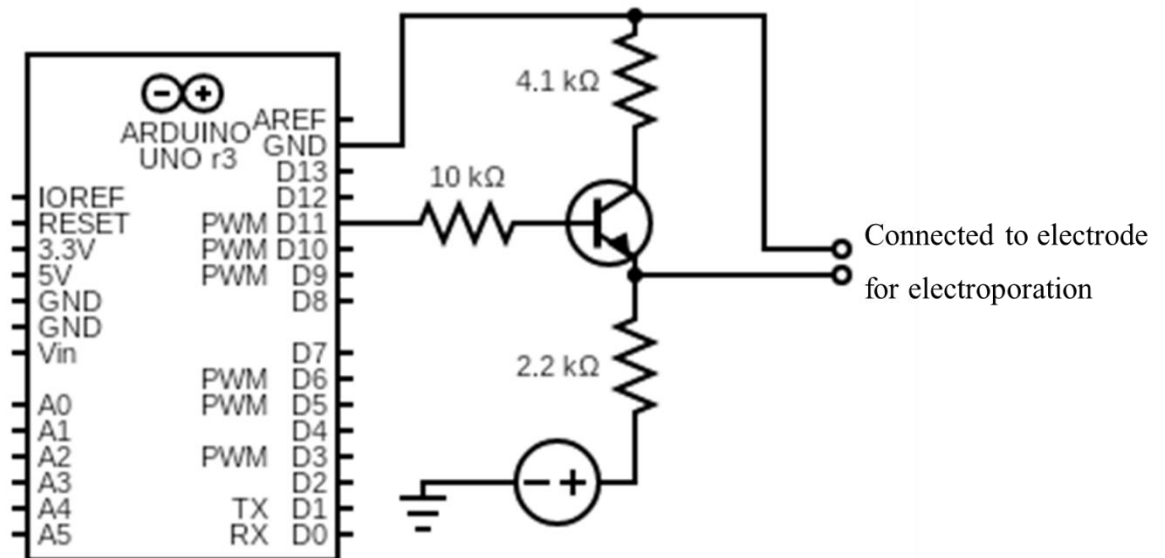
15. For the circuit connections, two platinum electrodes are connected to the output port.



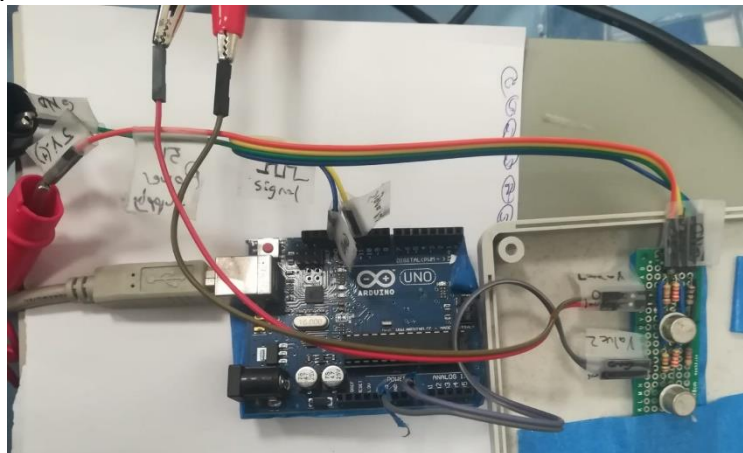
Red indicates the positive electrode and green representing the negative electrode. These wires are connected to the resistance board shown below.



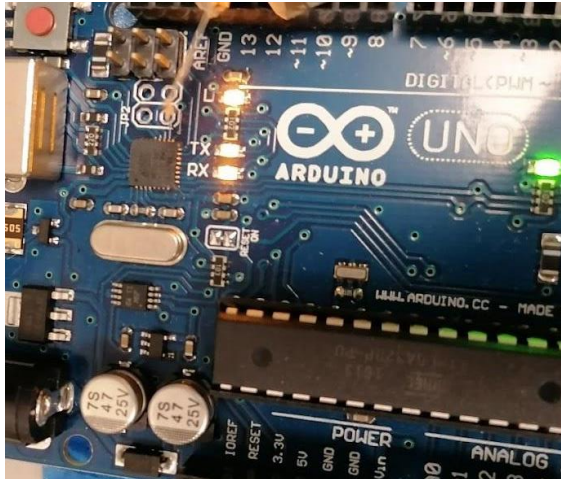
The circuit diagram is as shown below:



16. The red and the brown wires on the board are connected to the electrodes.
17. The gray wires are connected to the ground of the arduino and the yellow and blue wires are connected to the 11th and 13th pins on the arduino board.
18. The orange and green wires from the board are connected to the power supply source, orange to the positive and green to the ground.
19. The resistors are as indicated in the image above.
20. The entire setup is as follows:



21. The Arduino is connected to the computer with a USB cable.
22. The board must have three lights, one TX, RX and L. The green light indicates that the board is connected to a power supply. All lights are lit up in the image below



23. Connect the power port to the power supply and turn on channel 2.
24. The output button on the power supply should be pressed and immediately after, the LabView program must be run using the indicated button



25. The binary light will start flashing indicating that the signals have been received when
26. Ensure that the power supply output voltage is set to 13.562. This voltage is experimentally optimized.
27. Ensure that the selected ports are the same as the connections made on the board.

Microinjection and *in ovo* electroporation

Last updated: 18.02.20 by Merve Çelik

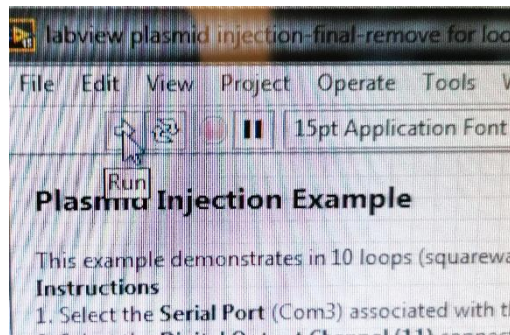
1) Materials and Equipment Used

- a) Stereomicroscope
- b) Micromanipulator
- c) External light source
- d) Forceps (WPI, 14099, Dumont #55)
- e) Microscissors (WPI, 14122, curved 5mm blades x 0.1 mm tips)
- f) Hamilton gastight syringe- removable needle type
- g) A 1.0 mm ID glass capillary tube (WPI, thin wall with filament, TW100F-5)
- h) Injection solution (Plasmid or siRNA)
- i) Chick embryo *in ovo* (White Leghorn)
- j) Egg holder
- k) 37°C chick ringer solution
- l) Electroporation part: Power source (5-20V), electronic chopper circuit (transistor circuit), Arduino for square wave generator, labview program to specify duty cycle.
- m) Electrodes: Platinum electrodes (125 micron diameter)

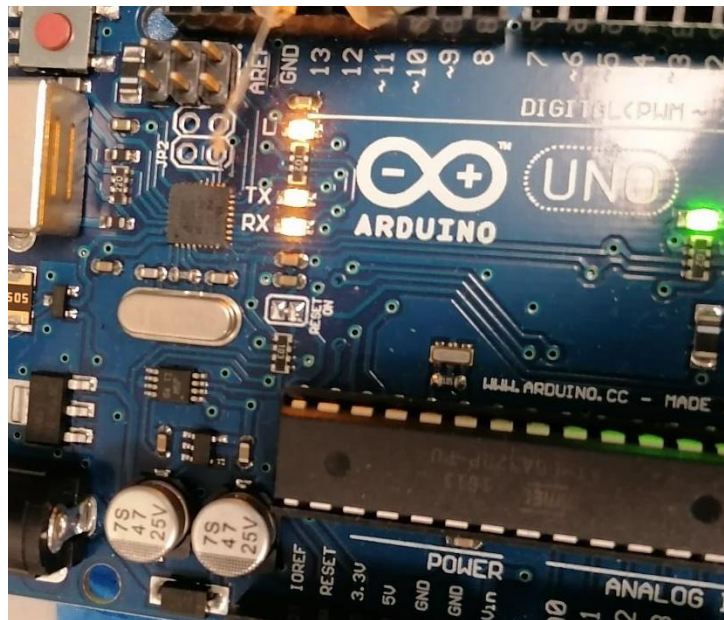
2) Method

1. Fertilized White Leghorn chicken eggs were incubated at 37.5°C and 65% relative humidity until HH18.
2. A small window was opened on the eggshell. The outer membrane is carefully removed under the stereo microscope. The inner membrane is also carefully removed under the stereomicroscope and should not form bleeding in the embryo.
3. The electrodes are cleaned with a 37-degree chick ringer solution. They are placed on the yolk. Care is taken that the electrodes do not come over any veins.
4. The glass capillary filled with the injection solution is approached and positioned. The entire egg is positioned under a stereomicroscope and the embryo is oriented to facilitate injection at the desired site. See figure 1.
5. About 2 µl of solution is injected into the region between right and left aortic arches.
6. Subsequently, a 37-degree chick ringer solution is dropped rapidly on the embryo.
7. Square wave electric current is applied at appropriate voltage, pulse and time. For the aortic arch region 14-15V, 10 pulses, 50 ms. Bubbles are formed in the electrodes when the electric current is applied. See figure 2.

- a. First, click the run button in the Labview program.



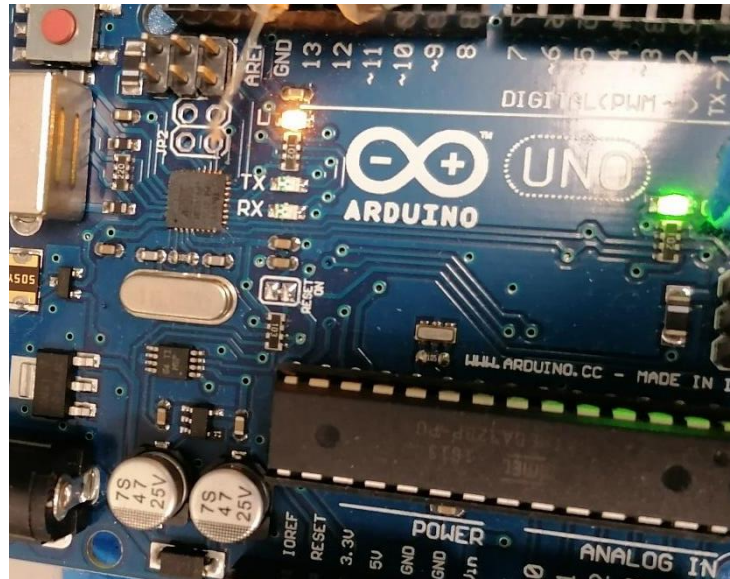
- b. As soon as the binary light is on in the electronic chopper circuit (the binary light will start flashing),



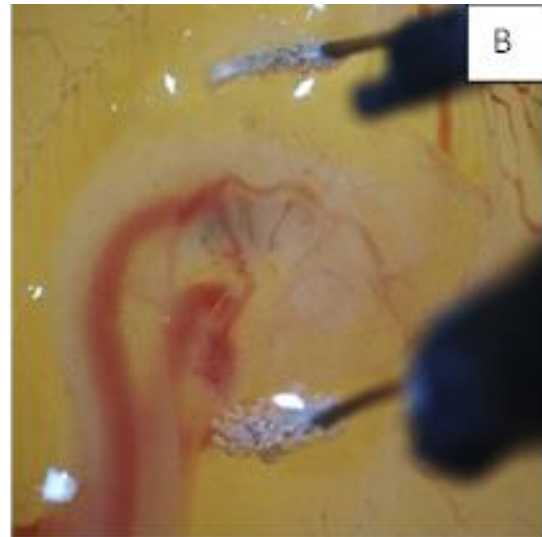
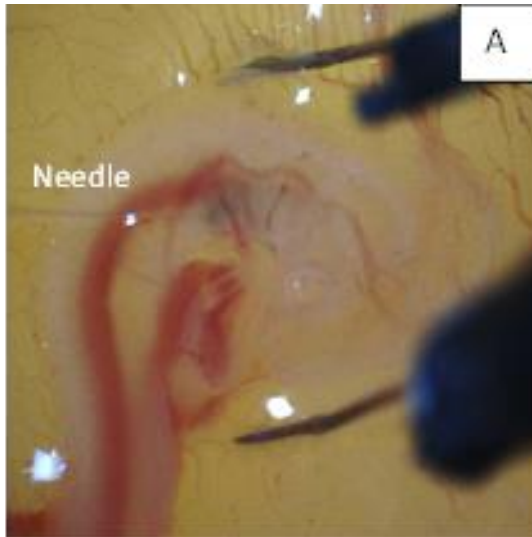
- c. Press the output button in the power source.



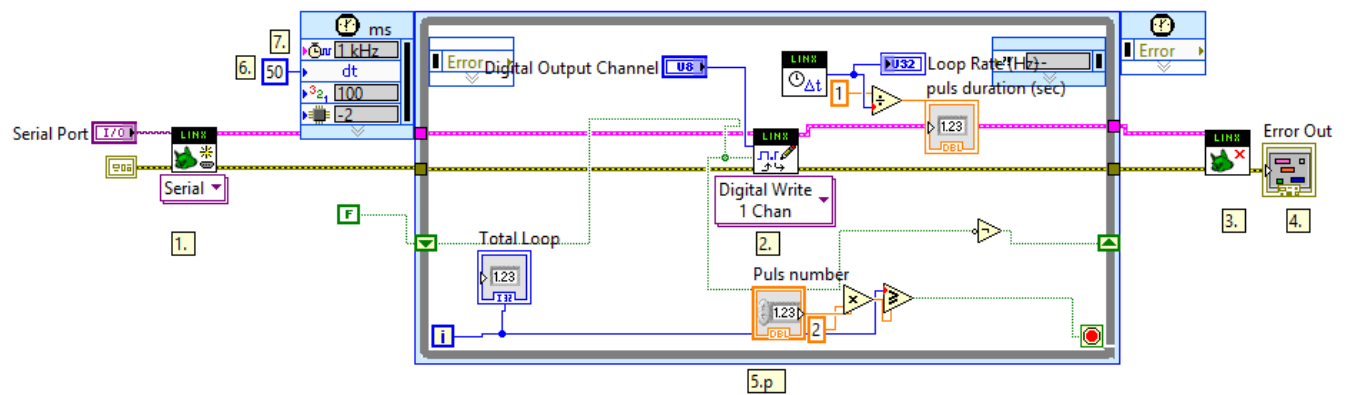
- d. The binary light in the system will start flashing.



8. Immediately thereafter, the 37-chick chick ringer solution is dropped into the embryo.
9. Then the egg is closed with parafilm and it will be incubated until HH21 or HH24.
10. After 24 hours later (or when the embryos reach the desired stage) the fluorescent tagged gene injected can be visualized on a confocal or fluorescent microscope.



Labview Design:



1. Open a connection to the LINX device.
2. Write the value to the specified DO channel.
3. Close the connection to the LINX device.
4. Handle Errors
5. Puls sayisi
6. 1 pulse duration- Frequency of one period (50msec.)
- 7.

Instructions

1. Select the **Serial Port (Com3)** associated with the LINX Device.
2. Select the **Digital Output Channel (11)** connected to the LED.
3. Click the **Run Arrow**.

